

Research Log

Tracking Medieval Manuscripts by Reverse Engineering the Paper-making Process
Zhanfeng Zhou | Robinson College | Supervisor: Dr Matteo Seita

1 Jun 2026

Submission day

- Submitted final report (38 pages, including title page and one mandatory risk-retrospective appendix), Technical Abstract as a standalone PDF, and this log book.
- Built final PDF locally using TinyTeX; verified 38 pages, no LaTeX errors. Replaced LaTeX-rendered cover with official CUED front page (Word template filled and exported to PDF).

31 May 2026

Reproducibility check, last polish

- Rebuilt PDF end-to-end and cross-checked every quoted number in Technical Abstract and §4 against the per-folio JSONs under `results/<folio>/`. All but two of the dataset 2 metric values reproduce.
- Noted residual data integrity issue: `results/2/fit_quality.json` and `results/2/interval_distribution.json` are byte-identical to the dataset 2b files (overwrite during archiving on 29 May), while `self_contrast.json` is from the original 18 May run. Recorded for viva preparation.
- Final typo pass across all chapters.

30 May 2026

Figure pass, duplication artefact discovery

- Generated the missing §1, §2 and §3 figures: folio anatomy close-up, DRM hardware schematic (TikZ), per-pixel DRP polar plot, anisotropy and azimuth maps, patchwise trig-mask, two-azimuth and four-azimuth grazing image panels, and the f5v overlay zoom.
- While generating the anisotropy map, observed a horizontal duplication artefact in the dataset 4 DRP cache at offset ≈ 1364 px. Cross-correlation peak of the cache mean is 0.64 at $(\pm 1364, 0)$; absent in any single (ϕ, θ) frame (peak < 0.18 and at a different offset). This is acquisition-level, not a pipeline bug.
- Cropped all DRP-derived figures to columns 0 to 2000 of the raw image so only one copy of the folio is visible. Updated captions accordingly.

29 May 2026

Cross-machine continuation

- Pulled main branch to home laptop. Python venv was broken (iCloud sync had renamed symlinks with spaces). Rebuilt venv, reinstalled numpy/scipy/matplotlib/opencv-python/pillow/PyYAML.
- First build of the report on this machine: installed TinyTeX, added biber, biblatex, titlesec, siunitx, microtype, cleveref, subcaption, pgf, caption via `tlmgr`. Build clean at 38 pages.

28 May 2026

Reference and §4 restructure

- Added seven literature references to §1 and §6.3: Bernstein and WIES watermark databases, Atanasiu AD751 software, Malzbender (PTM), Sethares (WIMSY), Sindel (ChainLineNet), Grossmann (spectral TV).
- Restructured §4 Benchmark from a metric-by-metric layout into DRP-first then MSI-second sections to mirror the detection chapter.
- Filled supervisor name and Robinson College on the front matter; resolved citation TODOs throughout.

27 May 2026

Body draft, first-round proofread

- Expanded skeleton from $\sim 2k$ to $\sim 9k$ words; six body chapters in place. Split the detection chapter into single-image MSI variant and multi-phi DRP variant.
- Added the pipeline diagram (TikZ block), aggregated spectrum figure for dataset 10, and per-image phase polar plot showing the π -flip polarity correction.
- Drafted the Technical Abstract from the headline numbers.

- Wrote the Risk Assessment Retrospective (mandatory ≤ 1 A4 appendix).
- Two proofread rounds. Fixed multiple broken cross-refs and track-attribution errors (multi-phi-only metrics being claimed on single-image MSI inputs).

26 May 2026

Report restructure

- Restructured report scaffold into Andrew Liang's 2024 narrative-chapter layout. Six chapters: Background, Initial Detection Attempts, Detection, Validation, Discussion, Conclusion.

25 May 2026

Manual GT annotation tool, report scaffold

- Built two-stage interactive annotation tool: bounding box selection (re-used `select_bbox.BBoxSelector`), then per-wire click placement with scroll-wheel zoom and live read-out of mean gap and lines/cm.
- Ran annotation on every folio of the nine-folio MSI benchmark. Saved as `manual_gt.json` per folio with both pixel gaps and the physical cm scale.
- Scaffolded the report directory under `report/`.

24 May 2026

Inter-counter variability hypothesis

- Spreadsheet density disagrees with pipeline on seven of nine MSI folios, errors -11% to -71% . Initial read was systematic detector bias.
- Re-read Da Rold ch. 1–2 on stock-vs-folio distinction. Hypothesis: spreadsheet records one folio per stock; within-stock density varies between folios.
- Discussed with supervisor: build per-folio manual ground truth instead of trusting the catalogue. Decision pivots the whole evaluation.

23 May 2026

Spectral-power line-direction detection

- Replaced projection-autocorrelation with a spectral-power criterion in `auto_detect_line_dir`. The autocorrelation was sensitive to the absolute amplitude of side-lobes and flipped sign on some folios.
- Added dual-band cross-validation: the peak energy in the candidate-period band must be stronger than in any neighbouring band of equal width.

22 May 2026

Synthetic phantom, spreadsheet comparison

- Built a synthetic phantom generator: Gaussian-comb intensity at a controlled period, orientation, wire σ and additive noise. Image generation runs in < 10 ms; pipeline runs in seconds.
- Four sweeps. Period 10–60 px, SNR 2–50, angle -30° – $+30^\circ$, sigma 1–8 px. Headlines: period recovery within $\pm 2.5\%$ across 10–50 px; essentially SNR-invariant down to SNR = 2; angle within 1° except at $+90^\circ$.
- Spreadsheet-comparison driver added; ran on f5v and f9v.
- Found that TX940IR images need `wire_is_darker=false` because transmitted-light contrast inverts the wire signature. Fixed and re-archived Kk1-5 results.

21 May 2026

Per-folio result archives

- Per-serial result archiving: each folio's outputs land at `results/<folio>/`, with the YAML config copied alongside so the run is replayable.
- Plain-language HTML report per run for sharing with the Manuscripts Lab.
- 1 cm scale grid overlay on the detected laid-line grid as a visual sanity check.

19 May 2026

MSI single-image pipeline

- Built the single-image MSI track: radial FFT of the column-mean $\rightarrow$ single Gabor cleanup pass $\rightarrow$ phase fit $\rightarrow$ grid. Bypasses the per-pixel direction map of the legacy track entirely.
- Added `detect_paper_roi` utility for MSI transmitted-light images: a band-pass texture filter separates paper from light-box frame, then a morphological close gives a clean rectangle.

18 May 2026

Half-period polarity correction

- Found per-image phase was systematically offset by exactly π between reflective and transmissive geometries. Wire shadow lands on the cosine maximum in reflective light but on the minimum under transmission.
- Auto-correction step in the multi-phi aggregation: examine whether the cosine fit's peak sits over a darker or brighter pixel; flip by half-period if needed.
- Loader cleanup: `crop_roi`, `period_range_px` and `wire_is_darker` added as proper Settings fields; stale-background detection with on-the-fly Gaussian-blur fallback.

17 May 2026

Re-engaging after Easter break

- Reorganised the codebase for re-engagement: stage-based subpackages (`stage0_loader`, `stage0_drp`, `stage1_features`, `stage2_enhance`, `stage3_detect`, `stage4_viz`, `stage5_evaluation`). Each stage advances the pipeline one step.
- Verified pipeline output unchanged at period 26 px, 18.2 lines/cm before and after the refactor.

12 May 2026

Plan for the Easter term

- Spent the past month on other Tripos modules and revision; project work on hold.
- Today: full read of own code. Bottlenecks identified: the per-pixel direction map at the front of the legacy pipeline carries most of the noise into the frequency stage and is the single biggest weakness.
- Plan for the next two weeks: (i) build a single-image FFT track that bypasses the direction map; (ii) keep the multi-phi idea but operate on raw grazing images, not the trig mask; (iii) characterise both on a synthetic phantom before touching real folios.

27 April 2026

Evaluation framework brainstorm

- Reading Otsu, Sezgin and Sankur reviews on the quality metrics used in periodic-texture detection. Most published methods report a single density figure with no confidence.
- Drafted on paper a five-metric bundle: harmonic-fit R^2 ; interval CV; self-contrast z -score against null directions; split-half stability across DRP partitions; per-segment wire FWHM IQR. Each is sensitive to one failure mode and largely insensitive to the others.
- Plan: implement once the detection track is rebuilt.

18 April 2026

Literature catch-up

- Hiary and Ng (2007) – projection-Fourier method on transmitted light. Closest published predecessor; user-supplied orientation is a recurring weakness.
- Grossmann et al. (2023) – spectral TV decomposition for chain and laid line extraction on reflected light. Produces enhanced images, not numbers.
- Gorske et al. (2021) – moldmate identification combining watermark geometry, chain-line spacing and laid-line density into one match score.
- Sethares et al. (2023) – WIMSY transmitted-light hardware.
- Sindel et al. (2021) – ChainLineNet, deep segmentation on transmitted-light scans.
- All five end up in the literature review of §1.2.

2 April 2026

End of Lent term

- Stepping away from the project until mid-Easter break. Module assessments and revision take priority for the next four weeks.
- Snapshot of current pipeline: DRP loader → per-pixel direction map (mag, deg) → patchwise trig mask → patchwise Gabor scan → overlay. Runs in ~ 4.5 min on a 480-image stack.
- Known issue carried over: half-period bias on phantoms.

26 March 2026

Patchwise Gabor wrap-up

- Locked the current Gabor-based pipeline as the end-of-Lent baseline. Patches of 512×512 , stride 256, periods 4–80 px.
- Observed half-period bias on synthetic-phantom-like inputs. Hypothesis: the absolute-valued Gabor response doubles the fundamental frequency; combined with the constant-Q (proportional bandwidth) scoring this drives the per-patch winner toward higher frequencies. Did not fix this term.
- Wrote up the issue in a notebook for the file.

14 March 2026

Meeting with Dr Seita

- Reviewed the patchwise Gabor pipeline at 4.5 min/folio, half-period bias.
- Discussion: before chasing more pipeline tweaks, design an evaluation framework so any future change can be measured. Quality metrics are more valuable than a single tighter density figure.
- Action items: (a) sketch metric bundle on paper; (b) synthetic phantom for ground-truth-level validation; (c) plan a multi-folio benchmark for next term.

2 March 2026

Reading: Gabor harmonic doubling

- Read on Gabor filter banks (Daugman, Bovik) and on the constant-Q vs constant-bandwidth distinction.
- Worked through on paper why $|G(x) * s(x)|^2$ doubles the fundamental of a sinusoidal $s(x)$. Confirmed the analytical source of the half-period bias.
- Next step ideas: replace patchwise Gabor with FFT peak detection (no squaring), or use a single Gabor with phase fit (no patch voting).

24 February 2026

Elevation angle selection

- Added `theta_min_deg` parameter to filter DRP slices to grazing-incidence only ($\theta \geq 30^\circ$ from page normal).
- Motivation: low- θ slices carry most of the directional laid-line signal; near-normal θ adds illumination noise without adding signal.
- Implementation: `apply_theta_min_filter` in `paperdrm/stage0_drp/slicing.py`.

21 February 2026

Hanning-weighted patch stitching

- Replaced hard patch tiling with Hanning-window overlap-weighted stitching. Removes patch-boundary discontinuities that were visible in the dominant-orientation map.
- Per-pixel result is now a smooth function of position rather than a piecewise-constant tile field.

16 February 2026

Lab visit: re-imaging dataset 4

- Re-imaged folio Kk.1.5 with the $1.0\times$ lens that arrived from the lab manager last week. Two scans at field of view ~ 8.65 cm.
- The flatter sample tray (manufactured to 2 mm thickness against the earlier 3 mm spec) is in. Less sample bow visible in the live preview.
- Cable tie to the camera frame in place; image stability checked.

11 February 2026

Patch-wise trigonometric map

- Tile the page with overlapping 512×512 patches; circular-mean orientation per patch.
- Each pixel receives a Gaussian likelihood of agreeing with its patch's dominant direction.
- Output is a greyscale image in which the laid-line orientation is locally enhanced relative to the per-pixel noise.

6 February 2026

Gaussian-orientation trigonometric filter

- Convert per-pixel azimuth map into a laid-line-likelihood greyscale. Gaussian centred on the target line direction with $\sigma = 10^\circ$; full 360° wrap-around distance.
- Implemented `azimuth_to_laidline_gray` in `paperdrm/stage2_enhance/trig_mask.py`.

3 February 2026

Gabor at fixed frequency for phase

- Fixed Gabor filter at the dominant frequency to infer wire phases (positions). Avoids re-scanning over period at the phase step.
- Output: 1D signal whose peaks coincide with detected wire positions.

6 December 2025

Literature review on visible-light imaging modalities

- Reflected light: easy to obtain, but low contrast on internal features, noise and ink interference.
 - Decompose images by scale and contrast; separate low-contrast structures from high-contrast ones (band-pass in scale space). Spectral TV decomposition is one realisation.
- Raking light: reveals surface topography.
- Transmitted light: cleaner internal-feature signal, complicates interpretation.
- Desire: build a model that maps laid line, chain line and watermark to a per-folio descriptor.
- Pipeline sketch: preprocessing $\rightarrow$ enhancement $\rightarrow$ detection $\rightarrow$ descriptor.

26 November 2025

Background substrate

- A4 paper does not work as the background under DRM. Surface is not actually even at the microscope scale and adds its own anisotropy.
- Need a flatter substrate.

24 November 2025

Lab visit – field of view

- Field of view is approximately 4×2.2 cm at the widest setting with the $1.3\times$ lens. Very small for a whole manuscript folio.
- Discuss with supervisor whether to step down magnification or to accept a per-patch stitching workflow.

17 November 2025

Meeting with Prof Da Rold

- Fetched one large and one small paper sample.
- Question: is the small sample at least as large as a real manuscript folio? Confirm with Prof Da Rold.
- Question: what is the difference between the two sides of a sample (felt vs wire)?

17 November 2025

Meeting with Dr Seita

Tray.

- Mount on the DRM platform with bolts.
- If ferromagnetic, use magnets to flatten the manuscript on top.
- Does tray thickness affect sample centricity? Light source emits near-parallel light; a 2 cm elevation in sample height causes up to $2/\tan(15^\circ) \approx 7.46$ cm of illumination-point shift – much too large. Need to measure the actual effect.
- Tray must be very flat and thick enough not to bend in handling.

Cable.

- The data cable behind the camera must be tied to the frame so it does not contact the sample during a scan.

Lens.

- Replace the 1.3× lens with the available 1.0× first. Report the camera make and model to Dr Seita so he can buy a 0.5× if needed.

Sample understanding.

- Watermarks and thickness features are more visible at high θ ; small surface features at low θ . Verify and explain.

31 October 2025

Paper reading

- Lins et al. – using paper texture for binarisation. Idea: model the surface as a vector in a parameter space; do matching by Euclidean distance.
- What pre-processing is needed before matching? Uniforming mean and variance?
- Toreini et al. – texture-based paper fingerprinting. They construct a 2-bit code per pixel from the Gabor response $C(x, y) = I(x, y) * G(x, y)$ and measure similarity in fractional Hamming distance.
- Note from the paper: surface alone misses non-even thickness, random impurities and different opacities. Motivates trying transmitted light if the project ever has access.
- Action: replicate Andrew Liang’s published output on a dataset with a recognisable feature.

20 October 2025

Group meeting

- Group themes: LRC-MRM imaging; Ti-6Al-4V surface characterisation. My project sits in the imaging side.

17 October 2025

First project meeting

- Project objectives agreed with Dr Seita: re-engineer the paper-making process by recovering mould geometry from non-invasive imaging of historical folios.
- Actions: participate in weekly group meetings, practice measuring samples on the DRM rig, read the relevant literature.